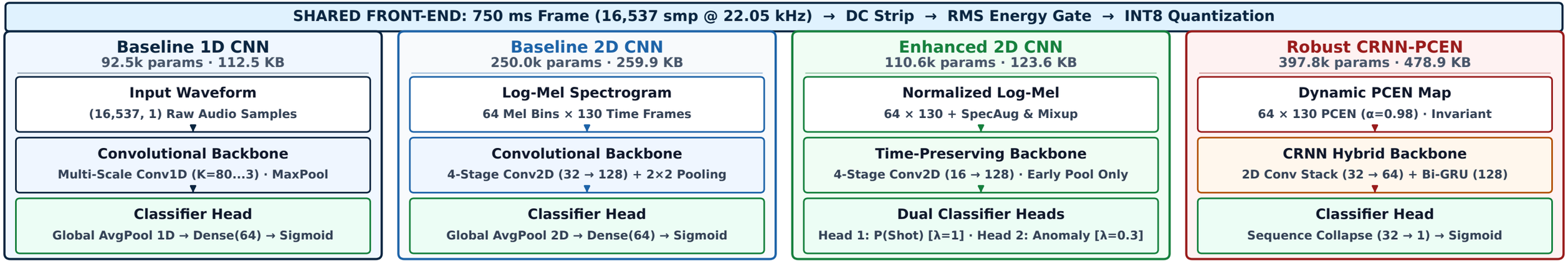
SHARED FRONT-END: 750 ms Frame (16,537 smp @ 22.05 kHz) → DC Strip → RMS Energy Gate → INT8 Quantization
Baseline 1D CNN
92.5k params · 112.5 KB
Input Waveform
(16,537, 1) Raw Audio Samples
Convolutional Backbone
Multi-Scale Conv1D (K=80...3) · MaxPool
Classifier Head
Global AvgPool 1D → Dense(64) → Sigmoid
Baseline 2D CNN
250.0k params · 259.9 KB
Log-Mel Spectrogram
64 Mel Bins × 130 Time Frames
Convolutional Backbone
4-Stage Conv2D (32 → 128) + 2×2 Pooling
Classifier Head
Global AvgPool 2D → Dense(64) → Sigmoid
Enhanced 2D CNN
110.6k params · 123.6 KB
Normalized Log-Mel
64 × 130 + SpecAug & Mixup
Time-Preserving Backbone
4-Stage Conv2D (16 → 128) · Early Pool Only
Dual Classifier Heads
Head 1: P(Shot) [λ=1] · Head 2: Anomaly [λ=0.3]
Robust CRNN-PCEN
397.8k params · 478.9 KB
Dynamic PCEN Map
64 × 130 PCEN (α=0.98) · Invariant
CRNN Hybrid Backbone
2D Conv Stack (32 → 64) + Bi-GRU (128)
Classifier Head
Sequence Collapse (32 → 1) → Sigmoid